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                                spectral garden 2026
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        CHOP / SCREW / MELANCHOLY AUDIO PROCESSING SYSTEM [v1.0.0]
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```

1. System Overview

d o w n e r is a real-time creative audio effect designed to recreate the characteristics of chopped & screwed tape deceleration, rhythmic stuttering, and atmospheric, low-fidelity environments.

Inspired by legacy console configuration menus and vintage tape degradation, the system combines:

- A latency-free **dual-tap pitch processor**
- An independent **vocal formant filter bank**
- A host-synchronized **stutter (chop) engine**
- A dynamic, **physical-modeling rain generator** that automatically gates according to the input signal's envelope to prevent background static when audio is paused.

2. User Manual (Consumer Guide)

2.1 Interface Overview

```
+-------------------------------------------------------------------------+
| [d o w n e r]  SYSTEM_BOOT // REAL_TIME_DECAY                           |
+-------------------------------------------------------------------------+
|  [ PRESET SELECTOR ]              ( MELT )     ( CHOP )    ( GLOOM )    |
|  01. Classic Screw                                                      |
|  02. Heavy Chop                     0.70x       0.50Hz       40%        |
|  03. Silent Hill Fog                                                    |
|  >04. Tape Stop Abyss                                                   |
|  05. Ethereal Ghost             [========================] +0.0 st      |
|  ...                            F O R M A N T                           |
|                                                                         |
|                                 [========================] 100% wet     |
```

```
|                                 M I X                                  |
+------------------------------------------------------------------------+
| MEMORY CARD SLOT [1] -- EXPANSION MODULE DETECTED                      |
+------------------------------------------------------------------------+
```

2.2 Parameter Guide

MELT (Rotary Knob)

- **Description:** Controls the overall playback rate and physical degradation of the signal.
- **Behavior:** Sweeps the pitch smoothly from original speed (1.0x) down to two octaves below (0.25x). Simultaneously blends in slow tape flutter (wow and flutter emulation at $1.4\ \text{Hz}$) to simulate aging physical tape heads.

CHOP (Rotary Knob)

- **Description:** Governs the frequency and depth of the rhythmic stutter loop.
- **Behavior:** Activating this control engages the sample-accurate loop-stutter engine. At higher values, the engine captures small, periodic subdivisions of incoming audio and loops them, mirroring traditional "chopped" physical production techniques.

GLOOM (Rotary Knob)

- **Description:** Controls environmental dampening, delay feedback space, and the rain generator mix.
- **Behavior:** Sweeps a steep low-pass filter down to $180\ \text{Hz}$ to remove high-frequency clarity. Simultaneously turns up a long, dark feedback delay path and mixes in the physical modeling rain soundscape.

FORMANT (Horizontal Fader)

- **Description:** Manually shifts or preserves vocal formants (throat size) independent of playback pitch.
- **Range:** $-12.0\ \text{st}$ to $+12.0\ \text{st}$ (semitones).
- **Behavior:** Helps counteract vocal pitch changes. Keeping the formant positive while pitching down preserves natural vocal character, whereas shifting it negative accentuates a heavy, deep-seated growl.

MIX (Horizontal Fader)

- **Description:** Linear crossfader controlling dry/wet balance.
- **Range:**

0%
(true bypass) to 100% (fully processed wet signal).

2.3 Preset Directory

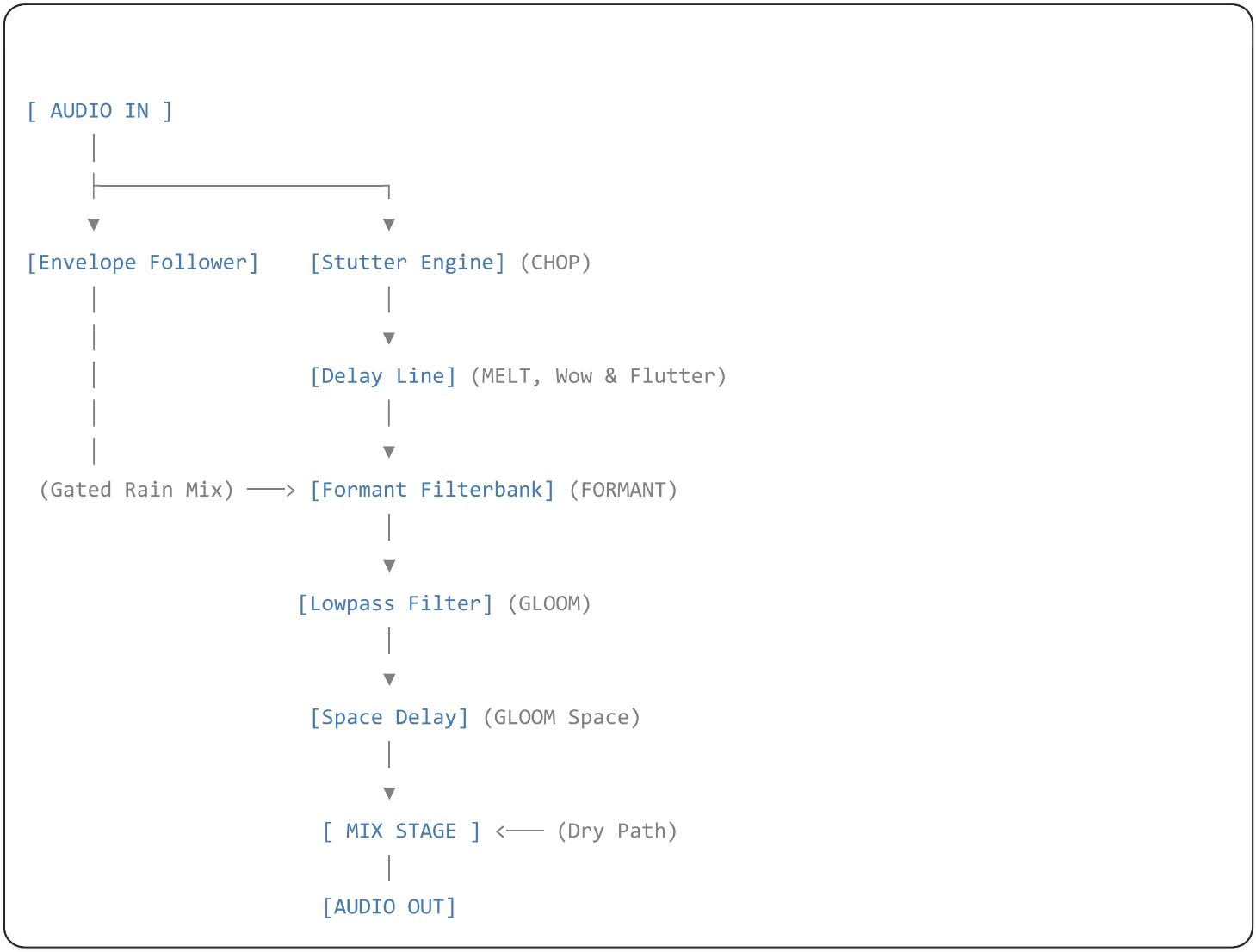
The plugin features 20 distinct profiles modeled after specific physical environments and vintage processing techniques, accessible via the interactive left-hand selector list:

Preset Name	Key Characteristics
01. Classic Screw	Standard tape rate deceleration with light flutter and 0 st formant shift.
02. Heavy Chop	Frequent, aggressive stuttering loops paired with moderate down-pitching.
03. Silent Hill Fog	Highly filtered tone, drowned out in long feedback echoes with elevated formant characteristics.
04. Tape Stop Abyss	Deeply pitched down (0.15x) with heavy stutter gating.
05. Ethereal Ghost	Pitched-down output paired with high formant shifting to generate hollow, disembodied textures.
06. Static Rain	Minimal pitch modification; isolates the dynamic physical modeling rain and low-pass filtering.
07. Melancholy Memory	A balanced blend of slow wow and flutter, moderate filtering, and soft echo.
08. Sub Zero Down	Deep pitch dropping (0.25x) with extreme low formant processing.
09. Liminal Corridor	Moderately dry setup focusing on mid-frequency filtering and space decay.
10. Corrupted Save	High chop values combined with extreme tape-stop transitions.
11. Dark Alley Echo	A wide, feedback-driven space with mild pitch shifting.
12. Screwed Vocal	Optimized down-pitching configured specifically to maintain vocal intelligibility.
13. Vinyl Dust	Light pitch wobble coupled with continuous, soft surface vinyl click impulses.
14. Slow Motion	Smooth, steady rate reduction without rhythmic stuttering.
15. VHS Degrade	Emulates heavily aged and worn consumer magnetic tape.
16. Rust & Decay	Severe low-pass filtering combined with heavy, slow pitch drift.
17. Graveyard Drift	A highly reverberant pitch shift with high formant preservation.
18. Under the Rain	Emphasizes maximum rain generation and dark space filtering.

19. Nostalgia Trip	Light, tape-style wobble with a moderate wet/dry mix.
20. Deep Void	High values on all parameters for absolute ambient processing.

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3. Technical Specification & DSP Architecture



3.1 Pitch Shifting (Dual-Tap Decelerator)

To avoid latency and window pre-ringing, the pitch-shifting engine does not rely on FFT or phase vocoder algorithms. It utilizes a continuous dual-tap circular delay line (

2.0 seconds
maximum duration):

- **Windowing:** Rather than basic linear ramps which cause phase cancellations and periodic clicks, the engine uses overlapping Hann Windows:

$$w(n) = 0.5 \cdot (1.0 - \cos(2\pi \cdot \phi))$$

- **Wow and Flutter:** A low-frequency oscillator ($1.4\ \text{Hz}$) modulates the delay read-head coordinates by up to $14\ \text{samples}$ scaled by the `Melt` parameter, simulating physical capstan tape slip.

3.2 Formant Filter Bank

Formants represent the resonant frequencies of the vocal tract. To preserve or shift them independently of pitch, the plugin routes the signal through a parallel/series 3-Band Biquad Filter Bank mapping the major human vocal resonances (F_1, F_2, F_3):

- **Center Frequencies:** $F_1 = 600\ \text{Hz}$, $F_2 = 1600\ \text{Hz}$, $F_3 = 2500\ \text{Hz}$.
- **Shifting:** The `Formant` parameter scales these center frequencies exponentially:

$$f_{\text{new}} = f_{\text{base}} \cdot 2^{\frac{\text{Formant}}{12}}$$

- **Implementation:** Second-order IIR Peak filters with constant-gain skirts are implemented to ensure resonance without digital overflow.

3.3 Rhythmic Stutter (Chop Engine)

The `Chop` parameter activates a sample-accurate loop capture buffer:

- **Capture Strategy:** Grabs incoming frames over a variable time window ($30\ \text{ms}$ to $300\ \text{ms}$) depending on the intensity of the control.
- **Pop Suppression:** To eliminate transient spikes and clicks at loop boundaries, a 150-sample linear crossfade window smoothly blends the end of the loop back into the start phase.

3.4 Dynamic Noise Gate & Physical Rain Modeling

To provide a rainy atmosphere without producing static background hiss when host playback is stopped or track audio pauses, d o w n e r implements an analog-style envelope follower:

- **Envelope Detector:** Uses a fast-attack ($7\ \text{ms}$) and slow-decay ($500\ \text{ms}$) peak rectifier.
- **Rain Generator:** White noise is filtered through an extremely high- Q bandpass while random click impulses simulate raindrops hitting hard surfaces.
- **Gating:** The volume of the rain generator is scaled directly by the calculated input envelope:

$$\text{Gain}_{\text{rain}} = \text{Gloom} \cdot \text{clamp}(0.0, 1.0, \text{Envelope} \cdot 18.0)$$

  This ensures that the rain fades out when the input track falls silent.

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4. Build & Compilation Instructions

4.1 Dependencies & Standards

- **JUCE Framework:** JUCE 8.0.14 (modules: `juce_audio_basics`, `juce_audio_processors`, `juce_gui_basics`, `juce_graphics`, `juce_core`).
- **C++ Standard:** C++20.
- **IDE/Compiler:** Microsoft Visual Studio 2026 (MSVC).

4.2 Projucer Configuration

1. Load the module requirements into your `.jucer` configuration.
2. Set the **C++ Language Standard** parameter to **C++20**.
3. Add an **MSVC Exporter** targeted at **Visual Studio 2026**.
4. Export and compile the generated `.sln` solution in **Release** configuration.

4.3 CMake Configuration

If building via CMake, configure your project target with the following setup:

```
cmake_minimum_required(VERSION 3.22)
project(downer VERSION 1.0.0)
```

```
find_package(JUCE CONFIG REQUIRED)

juce_add_plugin(downer
    COMPANY_NAME "DownerDSP"
    PLUGIN_NAME "downer"
    FORMATS VST3 AU Standalone
    PRODUCT_NAME "downer"
    IS_SYNTH FALSE
    NEEDS_MIDI_INPUT FALSE
    NEEDS_MIDI_OUTPUT FALSE
)

target_compile_features(downer PRIVATE cxx_std_20)

target_sources(downer PRIVATE
    Source/PluginProcessor.h
    Source/PluginProcessor.cpp
    Source/PluginEditor.h
    Source/PluginEditor.cpp
)

target_link_libraries(downer PRIVATE
    juce::juce_audio_processors
    juce::juce_audio_basics
    juce::juce_gui_basics
    juce::juce_graphics
    juce::juce_data_structures
    juce::juce_core
)
```